



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2005

Yearly Examination

YEAR 11

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 80 Marks

- Attempt Questions 1 - 4
- All questions are of equal value.

Examiner: *P. Bigelow*

Total marks – 80
Attempt Questions 1 - 4
All questions are of equal value

Answer each SECTION in a SEPARATE writing booklet.

Section A		Marks
Question 1 (20 marks)		
(a)	Express 0.0407 in scientific notation	1
(b)	If $\sqrt{54} = a\sqrt{6}$ find a	1
(c)	If $f(x) = 4x - 7$, find $f(a + 1)$	1
(d)	Solve	
(i)	$\frac{2x}{3} = -4$	1
(ii)	$2x(x - 4) = 0$	2
(iii)	$\frac{4}{y - 2} = 3$	1
(e)	If $\cos A = 0.407$ find A , where A is an acute angle.	2
(f)	Evaluate $\frac{4 \cdot 2^3 - 6 \cdot 1^2}{\sqrt{4 \cdot 6} - 0.3}$ correct to 1 decimal place.	1
(g)	Express $\frac{2}{\sqrt{3} - 1}$ with a rational denominator in <i>simplest</i> form.	2
(h)	Factorise fully:	
(i)	$2a^2 - 8$	2
(ii)	$2a^2 - 2a + ac - c$	2
(iii)	$4y^2 - 15y - 4$	2
(i)	Express $0.\dot{3}\dot{6}$ as a fraction in <i>simplest</i> form.	2

Section B (Use a SEPARATE writing booklet)**Marks****Question 2** (20 marks)

- (a) Find
- (i) $\lim_{x \rightarrow 4} \frac{x^2 + x - 20}{x - 4}$ 1
- (ii) $\lim_{x \rightarrow \infty} \frac{4 - x + 2x^2}{x^2 + 7x - 9}$ 1
- (b) Find x in the following:
- (i) $8^x = 2^{x-4}$ 1
- (ii) $\log_x 36 = 2$ 1
- (iii) $\log_3 \frac{\sqrt{3}}{9} = x$ 1
- (c) Differentiate the following
- (i) $y = (7 - 2x)^6$ 1
- (ii) $y = 7x^{-3} + 4x - 17$ 1
- (iii) $f(x) = x\sqrt{x-2}$ 2
- (iv) $f(x) = \frac{x+4}{x+5}$ 2
- (d) Write down a quadratic equation with roots 4 and -7 . 2
- (e) If $f(x) = \sqrt{x^2 + 16}$ find $f'(3)$. 2
- (f) Find the equation of the tangent to $y = x^2 + 3x + 4$ at the point where $x = 0$. 2
- (g) For the parabola $(x + 2)^2 = 4y - 8$, find the:
- (i) coordinates of the vertex; 1
- (ii) coordinates of the focus; 1
- (iii) equation of the directrix. 1

Section C (Use a SEPARATE writing booklet)**Marks****Question 3** (20 marks)

- | | | |
|-------|--|---|
| (a) | Solve the following pair of simultaneous equations | 3 |
| | $2x - y - 7 = 0$ $3x + 2y - 14 = 0$ | |
| (b) | Expand and simplify $(a + 1)^2 - (a + 1)(a - 1)$ | 3 |
| (c) | Use the quadratic formula to solve $x^2 + 4x - 6 = 0$.
Express your answers in <i>simplest surd</i> form. | 2 |
| (d) | Sketch the following on separate number planes: | |
| (i) | $3x - 2y + 12 = 0$; | 1 |
| (ii) | $y = \frac{4}{x}$; | 1 |
| (iii) | $y = x^2 - 4x + 3$. | 1 |
| (e) | Write down the exact values of: | |
| (i) | $\tan 30^\circ$; | 1 |
| (ii) | $\cos 210^\circ$. | 1 |
| (f) | Find to the nearest degree, the angle of depression of a boat 300 metres out to sea, from the top of a vertical cliff of height 50 metres. | 3 |
| (g) | If $5^x = 7$ find x correct to three decimal places. | 2 |
| (h) | Solve $2 \sin x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$ | 2 |

Section D (Use a SEPARATE writing booklet)**Marks****Question 4** (20 marks)

- (a) State the domain of the following:

(i) $y = \frac{2}{x^2 - 1}$ 1

(ii) $y = \sqrt{2 - x}$ 1

- (b) State whether the following are ODD, EVEN or NEITHER.
-
- Justify your answer. 6

(i) $f(x) = 1 - x^2$

(ii) $f(x) = \frac{x}{1 + x^2}$

(iii) $f(x) = -3x - x^2$

(iv) $f(x) = -3|x|$

- (c) Find the centre and radius of the circle 3

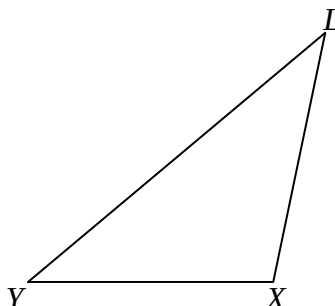
$$x^2 + y^2 - 4x + 6y - 3 = 0$$

- (d) (i) Graph the
- intersection*
- of the following regions on a number plane: 3

$$y \geq 2x + 6, \quad x \geq -3, \quad y < 8$$

- (ii) Find the area of this region. 1

- (e) A ship at
- X
- is 20 nautical miles from a lighthouse
- L
- , which is on a bearing of
- 025°T
- . The ship then sails due west to
- Y
- , from which the bearing of the lighthouse is
- 055°T
- .



Copy the diagram to your answer booklet.

- (i) Show that
- $\angle XLY = 30^\circ$
- . 2

- (ii) Show that
- $XY = 10 \operatorname{cosec} 35^\circ$
- and hence find the distance sailed in nautical miles, correct to 1 decimal place. 3

End of paper



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Sample Solutions

Marker AW

QUESTION 12005 2U Mathematics - Solutions

a) $0.0407 = \underline{4.07 \times 10^{-2}}$

b) $\sqrt{54} = 3\sqrt{6} \therefore \underline{a=3}$

c) $f(a+1) = 4(a+1) - 7$
 $\underline{= 4a - 3}$

d) i) $\frac{2x}{3} = -4$

$2x = -12$

$\underline{x = -6}$

ii) $2x(x-4) = 0$

$\underline{x=0 \text{ or } x=4}$

iii) $\frac{4}{y-2} = 3$

$4 = 3(y-2)$

$4 = 3y - 6$

$\underline{y = \frac{10}{3} = 3\frac{1}{3}}$

e) $\cos A = 0.407$
 $\therefore \underline{A = 65^\circ 59'}$

f) $\frac{4 \cdot 2^3 - 6 \cdot 1^2}{\sqrt{4 \cdot 6} - 0.3} = \underline{20.0}$

g) $\frac{2}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{2\sqrt{3}+2}{3-1}$
 $\underline{= \sqrt{3}+1}$

h) i) $2a^3 - 8 = 2(a^2 - 4)$
 $\underline{= 2(a+2)(a-2)}$

ii) $2a^2 - 2a + ac - c$
 $= 2a(a-1) + c(a-1)$
 $\underline{= (2a+c)(a-1)}$

h) iii) $4y^2 - 15y + 4$

$= 4y^2 - 16y + y + 4$

$= 4y(y-4) + (y-4)$

$\underline{= (4y+1)(y-4)}$

i) let $x = 0.36\dot{3}\dot{6}$ — ①

$100x = 36.36\dot{3}\dot{6}$ — ②

$99x = 36$ — ② - ①

$\underline{x = \frac{36}{99} = \frac{12}{33}}$

QUESTION 2

$$\begin{aligned} \text{a) i) } \lim_{x \rightarrow 4} \frac{x^2 + x - 20}{x - 4} \\ = \lim_{x \rightarrow 4} \frac{(x+5)(x-4)}{(x-4)} \\ = \underline{\underline{9}} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{ii) } \lim_{x \rightarrow \infty} \frac{4 - x + 2x^2}{x^2 + 7x - 9} \\ = \lim_{x \rightarrow \infty} \left[\frac{\frac{4}{x^2} - \frac{x}{x^2} + \frac{2x^2}{x^2}}{\frac{x^2}{x^2} - \frac{7x}{x^2} - \frac{9}{x^2}} \right] \\ = \lim_{x \rightarrow \infty} \left[\frac{\frac{4}{x^2} - \frac{1}{x} + 2}{1 - \frac{7}{x} - \frac{9}{x^2}} \right] \quad \textcircled{1} \end{aligned}$$

NOTE

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

$$= \frac{2}{1} = \underline{\underline{2}}$$

$$\begin{aligned} \text{b) i) } 8^x &= 2^{x-4} \\ \circ \quad 2^{3x} &= 2^{x-4} \\ \circ \quad 3x &= x-4 \\ \underline{\underline{x &= -2}} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{ii) } \log_x 36 &= 2 \\ \circ \quad x^2 &= 36 \\ \circ \quad \underline{\underline{x = \pm 6}} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{iii) } \log_3 \frac{\sqrt{3}}{9} &= x \\ \log_3 \sqrt{3} - \log_3 9 &= x \\ \frac{1}{2} - 2 &= x \\ \underline{\underline{x = -1\frac{1}{2}}} \quad \textcircled{1} \end{aligned}$$

$$\text{c) i) } \frac{dy}{dx} = -12(7-2x)^5$$

$$\text{ii) } \frac{dy}{dx} = -21x^{-4} + 4$$

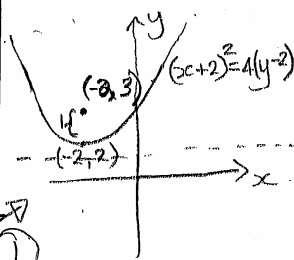
$$\begin{aligned} \text{iii) } f'(x) &= u'v + uv' \\ &= (x-2)^{1/2} + x \left(\frac{1}{2}(x-2)^{-1/2} \right) \\ &= \sqrt{x-2} + \frac{x}{2\sqrt{x-2}} \quad \textcircled{2} \end{aligned}$$

$$\begin{aligned} \text{iv) } f'(x) &= \frac{vu' - uv'}{v^2} \\ &= \frac{(x+5) - (x+4)}{(x+5)^2} \\ &= \underline{\underline{\frac{1}{(x+5)^2}}} \quad \textcircled{2} \end{aligned}$$

$$\text{d) } (x-4)(x+7) = \underline{\underline{x^2 + 3x - 28}} \quad \textcircled{2}$$

$$\begin{aligned} \text{e) } f'(x) &= \frac{1}{2}(x^2+16)^{-1/2} \cdot 2x \\ &= \frac{x}{\sqrt{x^2+16}} \quad \textcircled{2} \\ f'(3) &= \underline{\underline{\frac{3}{5}}} \end{aligned}$$

$$\begin{aligned} \text{f) } \frac{dy}{dx} &= 2x+3 \\ \text{when } x=0 \\ \underline{\underline{\frac{dy}{dx} = 3}} \quad \textcircled{2} \end{aligned}$$



$$\text{g) i) } (-2, 2) \quad \textcircled{1}$$

$$\text{ii) } (-2, 3) \quad \textcircled{1}$$

$$\text{iii) } y=1 \quad \textcircled{1}$$

★ Gradient tangent = 3.
when $x=0$ $y=4$.
 $4-4 = 3(x-0) \quad \& \quad 3x - y + 4 = 0$

QUESTION 3:

a) $2x - y - 7 = 0$ — ①

$3x + 2y - 14 = 0$ — ②

$2 \times ①$
 $4x - 2y - 14 = 0$ — ③

③ + ②

$$7x - 28 = 0$$

$$x = +4$$

∴ $y = 2x - 7$

$$y = +1$$

Solution is $x=4, y=1$

b) $(a+1)^2 - (a+1)(a-1)$

$$= a^2 + 2a + 1 - a^2 + 1$$

$$= 2a + 2$$

$$= 2(a+1)$$

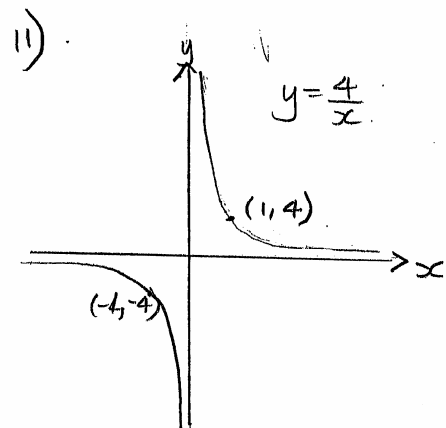
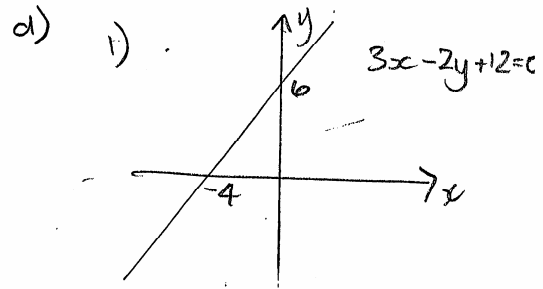
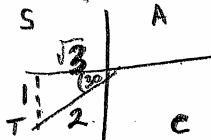
c) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$= \frac{-4 \pm \sqrt{16 + 24}}{2}$$

$$= \frac{-2 \pm \sqrt{10}}{1}$$

d) i) $\tan 30^\circ = \frac{1}{\sqrt{3}}$

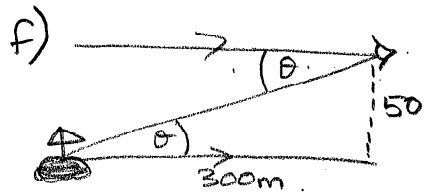
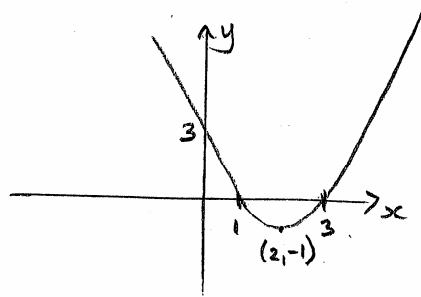
ii) $\cos 210^\circ = -\frac{\sqrt{3}}{2}$



iii)

$$y = x^2 - 4x + 3$$

$$y = (x+1)(x-3)$$



$$\tan \theta = \frac{50}{300} = \frac{5}{3} = \frac{1}{6}$$

$$\theta = 9^\circ 27' = 9^\circ \text{ (nearest degree)}$$

QUESTION 3 (CONT)

$$g) 5^x = 7$$

$$\log_5 7 = x$$

$$\therefore x = \frac{\log_{10} 7}{\log_{10} 5}$$

$$= 1.209 \text{ (3 dp)}$$

$$h) 2 \sin x + 1 = 0$$

$$\sin x = -\frac{1}{2}$$

$$x = 210^\circ, 330^\circ$$

QUESTION 4

a) i) $x \in \mathbb{R} : x \neq 1$

ii) $x \in \mathbb{R} : x < 2$

b) i) $f(x) = 1 - x^2$

$$f(x) = 1 - (-x^2)$$

$$= f(x)$$

∴ EVEN FUNCTION

ii) $f(x) = \frac{x}{1+x^2}$

$$f(-x) = \frac{-x}{1+(-x)^2}$$

$$= -f(x)$$

∴ ODD FUNCTION

iii) $f(x) = -3x - x^2$

$$f(-x) = 3x - x^2$$

∴ NEITHER

iv) $f(x) = -3|x|$

$$f(-x) = -3|-x|$$

$$= f(x)$$

∴ EVEN FUNCTION

c) $x^2 + y^2 - 4x + by - 3 = 0$

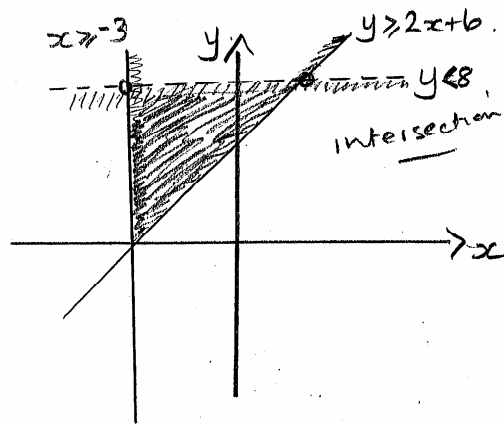
$$x^2 - 4x + (2)^2 + y^2 + by + 3^2 = 3 + 4 + 9$$

$$(x-2)^2 + (y+3)^2 = 16$$

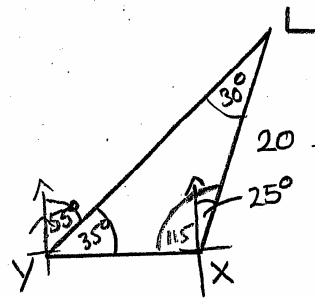
Centre of Circle $(2, -3)$

$$\text{Radius} = 4$$

d)



e)



i) $90^\circ - 55^\circ = 35^\circ$ (complement $\angle$ s)

ii) $\frac{XY}{\sin \angle LX} = \frac{LX}{\sin \angle YX}$

$$\frac{XY}{\sin 30^\circ} = \frac{20}{\sin 35^\circ}$$

$$XY = \frac{1}{2}(20) \operatorname{cosec} 35^\circ$$

$$= 10 \operatorname{cosec} 35^\circ$$

$$XY = 17.4 \text{ nautical miles}$$